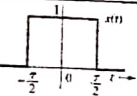


THIRD SEMESTER – B.TECH
END-SEMESTER EXAMINATION, NOVEMBER 2025

Course Code- EAEPC302
Course Title- Signal and Systems
Time- 3 Hour

Max. Marks- 50

Note: - Attempt all questions. Missing data/ information if any, maybe suitably assumed & mentioned in the answer.

Q No.	Question	Marks	CO
1a	Determine the power and Energy of the following signals: i) $x(t) = e^{-2t}u(t) + e^{2t}u(-t)$ ii) $y(n) = (1/5)^n u(n)$	5	CO 1, 2
1b	Determine whether the following signals are periodic, if yes find the fundamental period. i) $x(t) = e^{j(\pi/2)t}u(t)$, ii) $x[n] = \cos(5\pi n/6) + \sin(\pi n/4)$	5	CO 1, 2
1c	What are invertible systems? Determine if each of the following systems is invertible. i) $x(t) \rightarrow x(t-4)$ ii) $y[n] = \cos[x[n]]$	5	CO 3, 4
2a	Input-output relationship for a system is given by $y(t) = t^2 x(t-1)$. Determine whether the system is linear and time invariant.	5	CO 1, 2, 3
2b	Let $x[n] = 4\delta[n] + 2\delta[n-1] - \delta[n-3]$ and $h[n] = 3\delta[n+1] + \delta[n-1]$. Compute the following convolutions: a) $y[n] = x[n] * h[n]$ b) $y[n] = x[n+3] * h[n]$ c) $y[n] = x[n] * h[n+2]$	5	CO 2, 5
2c	Determine the convolution of the following two signals $x(t) = \begin{cases} 1, & 0 < t < T \\ 0, & \text{otherwise} \end{cases}$ $h(t) = \begin{cases} t, & 0 < t < 2T \\ 0, & \text{otherwise} \end{cases}$	5	CO 3, 4
3a	Find out the Fourier series representation for the given signal 	5	CO 2, 5
3b	Find the Fourier transform of the given signal $x(t) = e^{-at}u(t)$, $a > 0$	5	CO 2
3c	 Find the Fourier transform.	5	CO 3, 4
4a	Perform convolution of the following signal using Laplace transform $x(t) = e^{-3t}u(t)$, $h(t) = u(t-3)$	5	CO 2
4b	Find out the inverse Laplace transform and whether the given system is stable and causal or not $H(s) = \frac{s-1}{(s+1)(s-2)}$	5	CO 4, 5

4c	Determine the Laplace transform of $x(t) = t e^{-at}u(t)$	5	CO 3, 4
5a	Determine the Z transform of given signal along with ROC. $x[n] = 7\left(\frac{1}{3}\right)^n u[n] - 6\left(\frac{1}{2}\right)^n u[n]$	5	CO 2, 4
5b	Find the inverse Z transform of the following signal $X(z) = \frac{3 - \frac{5}{8}z^{-1}}{(1 - \frac{1}{4}z^{-1})(1 - \frac{1}{2}z^{-1})}, z > \frac{1}{2}$	5	CO 2, 4
5c	Determine the system function for the causal LTI system with difference equation $y[n] - \frac{1}{2}y[n-1] + \frac{1}{4}y[n-2] = x[n]$	5	CO 3, 4